

SV-16 microcontroller — PCB design blueprint

Status: design specification for review — no board has been built or ordered. Nothing in this document has been validated on hardware; every number is either taken from a cited document (Lattice datasheet / technical note / BSDL model) or measured from this repository's own build artifacts. Items that still need bench or vendor verification are collected in §13.2.

What it is: a complete component list, electrical specification and connection diagram for turning the SV-16 soft-core that lives in this repository into a **stand-alone MCU-class board** built around a single Lattice **LFE5U-12F-6TG144C** (ECP5, 144-lead TQFP, speed grade 6, commercial temperature).

It is not: a copy of an evaluation kit, and not a general-purpose FPGA dev board. The board exists to run the firmware in `firmware/` — boot from non-volatile memory, talk on a serial console, be reprogrammed in the field, and drive a small motor — which is exactly what the SoC in `rtl/` implements.

Device	LFE5U-12F-6TG144C — ECP5, 12,144 LUT4, 32 × sysMEM (72 KB), 28 multipliers, 2 PLL, 144-TQFP
Fit of the current SoC	9,191 LUT4 (75.7 %), 4,772 FF (39.3 %), 18/32 DP16KD (56.2 %), 1/2 EHXPLLL, 52/98 I/O
Bitstream	292,752 B compressed (<code>build/pll/sv16_top.bit</code>), 45.45 MHz max core clock
I/O banks used	0, 1, 2, 3, 7, 8 — all LVCMOS33 (3.3 V), 52 signals constrained, 46 spare I/O
Storage	2 × 3.3 V SPI NOR: one for the FPGA bitstream, one for SV-16 firmware images
Console / update	USB-C → CH340C → 115200 8N1 UART → ROM monitor (erase / upload / verify / boot)
Debug / config	1×10 0.1" and 2×5 0.05" JTAG headers; TCK ≤ 25 MHz (BSDL <code>TAP_SCAN_CLOCK</code>)
Power in	5–12 V DC barrel or USB-C 5 V; 3.3 V / 2.5 V / 1.1 V rails generated on board

0.1 Where the numbers come from

Fact	Source
Pin numbers, banks, pad names, the 98 user-I/O pins	<code>board/ecp5_tqfp144.json</code> , cross-built from the prjtrellis device database for LFE5U-12F TQFP144 (98 user-I/O pin numbers + banks) and the Lattice BSDL model FPGA-MD-02097 (LFE5U_25F_XXTG144, pad names, the 7 dedicated sysCONFIG pins, the 4 TAP pins, every VCC/VCCAUX/VCCIO/VSS pin)
Which pin carries which design signal	<code>constraints/ecp5_144tqfp.lpf</code> — the same file the bitstream is built with
Rail voltages, tolerances, ramp rates, POR trip points, LVCMOS33 range	ECP5 and ECP5-5G.pdf in this repository — FPGA-DS-02012-3.4 , Tables 3.1–3.5, 3.8, 3.11, §4.3.2
Configuration strapping, pull-ups, MCLK, PERSISTENT	Lattice TN1260 (ECP5 sysCONFIG) and TN-02039 (sysCONFIG usage guide); pull-up values from the ECP5 hardware checklist TN-02038-2.0 Table 6.2
JTAG header pinout	Lattice ECP5 Versa EB98/EB103 Appendix A (1×10 header) and the Digilent/FT2232H 2×5 0.05" standard

Fact	Source
Pinout of every constrained signal on the board	board/TQFP144_PINOUT.md (generated)

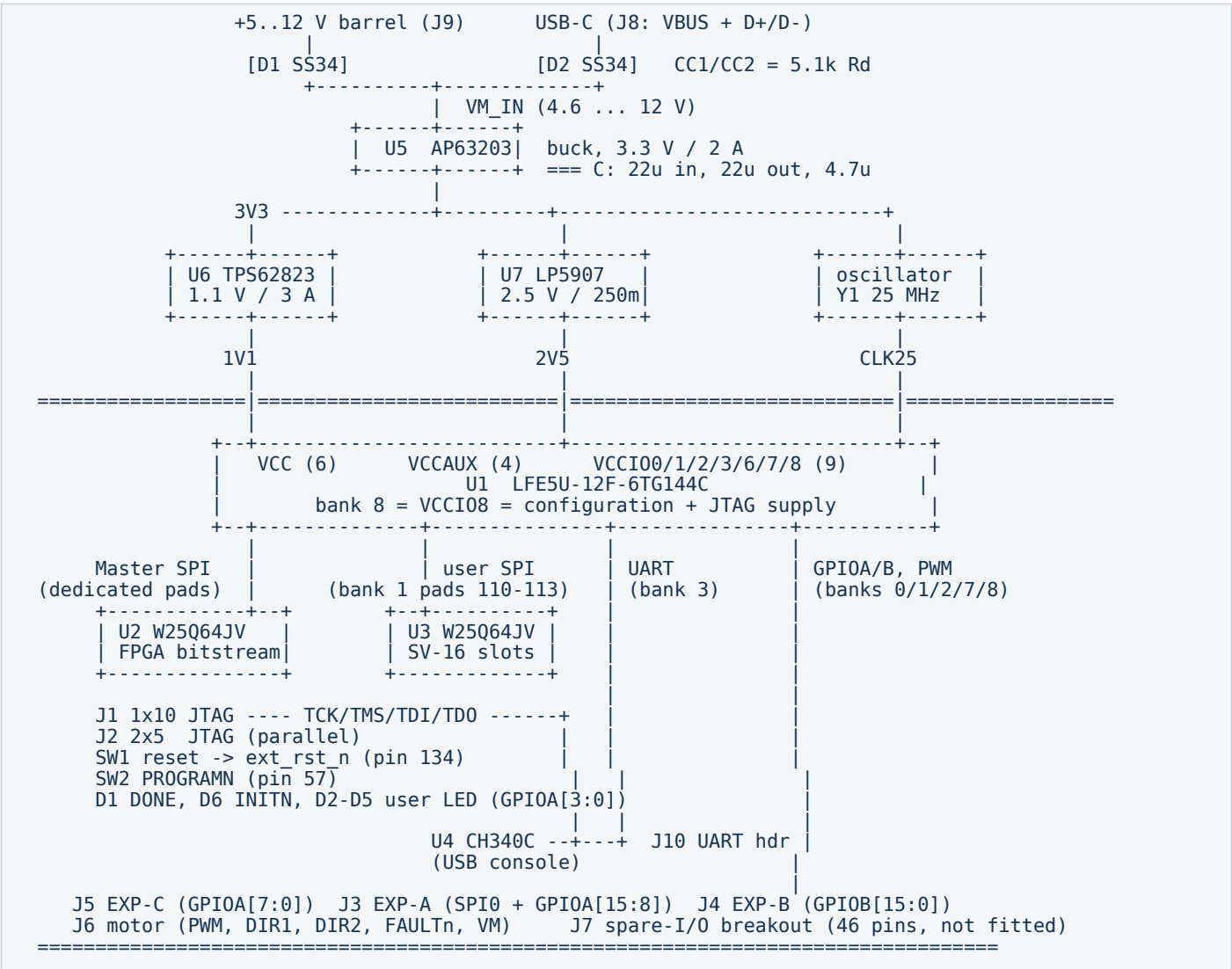
Printables. This document exists twice: BOARD.md (the source of truth, with the generated netlist appendix in board/TQFP144_PINOUT.md) and BOARD.pdf — the same content typeset for printing, with the ASCII diagrams kept in a monospaced face. Rebuild the PDF with `make board-pdf` (python3 -m pip install markdown xhtml2pdf once; nothing else in the build needs those packages).

Drift control. board/TQFP144_PINOUT.md is generated from the two files above and is verified in `make lint`:

```
scripts/sv16_board_pins.py      # regenerate the pin/net table
scripts/sv16_board_pins.py --check  # fail if the checked-in table is stale
scripts/sv16_board_pins.py --summary # bank/usage summary only
```

If somebody moves a signal in `constraints/ecp5_144tqfp.lpf`, `--check` fails until the board document is regenerated, so the schematic and the bitstream cannot silently disagree.

1. Board-level block diagram



The soft-core sees this board through its memory map: GPIOA/GPIOB at 0xF010/0xF070, PWM+quadrature motor block at 0xF030, SPI0 at 0xF050, UART0 at 0xF000, watchdog at 0xF080, SPI flash controller at 0xF0A0 (docs/MEMORY_MAP.md, docs/PERIPHERALS.md). Every board signal below maps to one of those registers — the board adds nothing the SoC cannot already drive.

2. Decisions that shaped the board (and why)

- 1. **One I/O rail: 3.3 V everywhere.** The existing LPF constrains all 52 signals to LVCMOS33, which needs VCCIO 3.135–3.465 V (datasheet Table 3.11). Rather than split banks across rails (which would force LPF and bitstream changes), all six used banks sit on 3V3. The flash, the USB bridge and the 25 MHz oscillator are all 3.3 V parts, so no level shifting is needed anywhere.
- 2. **Power-up order is guaranteed by construction, not by a sequencer.** In Master SPI mode the datasheet requires VCCIO8 to come up before (or together with) VCC/VCCAUX, or else PROGRAMN/INITN must be held low until VCCIO8 is valid (§3.5). The board derives 2.5 V and 1.1 V from the 3.3 V rail, so VCCIO8 is by definition the first rail to become valid (§4.3).
- 3. **Two SPI flashes, not one.** The FPGA's configuration port is not memory-mapped: the MSPI pads (CCLK = pin 54, MOSI = pin 47, MISO = pin 46, CSSPIN = pin 49) belong to the configuration logic while PERSISTENT is set, and the soft-core cannot read or write the bitstream flash through them. The SV-16 firmware store must therefore be a separate device on ordinary user I/O (bank 1, pins 110–113), which is what sv16_flash_ctrl and the bootloader already expect. Consequences and the pin-sharing rule are spelled out in §6.4 — this is the single most important thing to understand about this board.
- 4. **25 MHz single-ended oscillator, PLL on-chip.** The design is built for clk_25m on pin 133 and the PLL recipe CLKI 2 / CLKFB 3 / CLKOP 16 → 37.5 MHz core (VCO 600 MHz), giving 45.45 MHz of margin. A 3.2 × 2.5 mm 4-pin CMOS XO is cheaper and more robust than a crystal on the FPGA's oscillator pad, and it also supports the CLKSRC=osc build (25 MHz, no PLL) unchanged.
- 5. **USB-UART instead of a second USB device controller.** The console/update path is the same UART the ROM monitor already speaks: a CH340C makes it a single USB-C cable. The optional DTR→PROGRAMN transistor lets a host script reconfigure the FPGA, but it is jumper-disableable because DTR polarity differs between host tools.
- 6. **JTAG is the primary configuration and recovery path.** A 1×10 0.1" header (Versa-compatible) and a 2×5 0.05" header (FT232H/HS2-compatible) are wired in parallel, so both a cheap FT232H adapter and a Digilent-style probe work. JTAG always works even when the configuration flash is blank or corrupt — that is the brick-recovery path.
- 7. **Motor interface on the pins the SoC already drives.** pwm_out (pin 88, 16 mA), motor_dir1/2 (GPIOA[4]/[5] mirrored to pins 89/102) and motor_fault_n (pin 103, pulled up) go to a keyed 1×6 header with the motor supply passed through, so an external driver board (or a future on-board driver, §14 V3) plugs straight in. No new logic.
- 8. **Every spare pin is brought out.** 46 of the 98 I/O pins are unused by the current design; they go to a 2×25 footprint J7 (not fitted by default) with interleaved grounds so the board can grow without a respin.

3. Bill of materials

Quantities are for one board. "Alt" gives a drop-in alternative; prices are indicative 1-off 2026 distributor bands for budgeting only (verify at order time). Passives are 0402/0603 X7R/X5R unless stated; electrolytics are not required anywhere.

3.1 Active devices

Ref	Part number	Specification	Package	Qty	Alt / notes
U1	LFE5U-12F-6TG144C	CECP5 FPGA, 12,144 LUT4, 72 KB sysMEM, 28 mult, 2 PLL, VCC 1.1 V, VCCAUX 2.5 V, speed 6, commercial 0...85 °C Tj	TQFP-144 20×20 mm 0.5 mm	1	the part this repository is pinned to; must be the C (commercial) and 6 (speed) suffix

Ref	Part number	Specification	Package	Qty	Alt / notes
U2	W25Q64JVSSIQ	64 Mbit (8 MB) 3.3 V SPI NOR, 104 MHz, 2.7–3.6 V, –40...85 °C	SOIC-8 208 mil	1	FPGA bitstream flash (Master SPI). W25Q32JVSSIQ (4 MB) is enough — the bitstream is 293 KB
U3	W25Q64JVSSIQ	same device, second chip	SOIC-8 208 mil	1	SV-16 application flash (slot A/B firmware). 1 MB (W25Q80JV) is already 16× more than needed
U4	CH340C	USB 2.0 full-speed ↔ UART bridge, 3.3 V, internal oscillator, no crystal	SOP-16	1	console + firmware upload. Alt: CP2102N-A02-GQFN24 (no crystal, 3.3 V), FT231X (SSOP-20)
U5	AP63203WU-7	synchronous buck, 3.8–32 V in, 3.3 V fixed, 2 A, 1.1 MHz	TSOT-26	1	3V3 rail. Alt: TPS54331D (needs a divider)
U6	TPS62823DLR	synchronous buck, 2.4–5.5 V in, adjustable, 3 A	SOT-583	1	1V1 core rail, fed from 3V3 (sets the power-up order, §4.3)
U7	LP5907MFX-2.5/NOPB	PMOS, 2.5 V, 250 mA, 2.2–5.5 V in, 6.5 µV rms noise	SOT-23-5	1	2V5 (VCCAUX) rail, fed from 3V3
Y1	ASEMB-25.000MHZ-25-T00	25.000 MHz XO, 3.3 V CMOS, ±50 ppm, 15 pF, 4-pin	3.2×2.5 mm	1	clk_25m. Alt: SiT8008BI-23-33E-25.00000
D1	LTST-C191KGKT	green LED, "CONFIGURED" indicator (driven through Q2)	0603	1	lit = DONE high = user mode
D2–D5	LTST-C191KGKT	user LEDs (mirror of GPIOA[3:0])	0603	4	active low, code already drives them (assign led = ~gpio_a_out[3:0])
D6	LTST-C191KRKT	red LED, INITN / config-error indicator	0603	1	optional (DNP) — lit = INITN low (SRAM clear / config error)
D7	LTST-C191KRKT	red LED, 3V3 power-good	0603	1	
D8, D9	SS34	Schottky, 40 V / 3 A, input ORing + reverse protection	SMA	2	barrel and USB-C inputs into VM_IN
Q1	2N7002	N-MOSFET, PROGRAMN pull-down from DTR (optional auto-reconfigure)	SOT-23	1	disabled by JP1; see §7.2

Ref	Part number	Specification	Package	Qty	Alt / notes
Q2	MMBT3904	NPN, DONE indicator driver (keeps the open-drain DONE pin's logic level clean)	SOT-23	1	see §7.2
Q4	MMBT3906	PNP, INITN indicator driver (optional — fit only if you want a visible config-error LED)	SOT-23	1	see §7.2
U8	USBLC6-2SC6	USB ESD protection (D+/D−)	SOT-23-6	1	
—	SN74LVC1T45 (optional)	level shifter, only if a 5 V UART adapter is used on J10	SOT-23-6	0	not fitted by default: J10 is 3.3 V-only

3.2 Connectors, switches, protection

Ref	Part number	Specification	Qty	Notes
J1	1×10 0.1" header, vertical	2.54 mm, JTAG/config, Versa pinout (see §8)	1	primary programming header
J2	2×5 0.05" keyed box header	1.27 mm, Digilent/FT2232H pinout	1	wired in parallel with J1's JTAG pins
J3	2×10 0.1" header	EXP-A: SPI0 + GPIOA[15:8] + 3V3/5V/GND	1	
J4	2×10 0.1" header	EXP-B: GPIOB[15:0] + 3V3/5V/GND	1	
J5	1×10 0.1" header	EXP-C: GPIOA[7:0] + 3V3/GND	1	pins 2/3/5/7 share nets with the config flash — see §6.4
J6	1×6 0.1" keyed header	VM, GND, PWM, DIR1, DIR2, FAULTn	1	keyed (polarised) to prevent reverse plugging
J7	2×25 2.54 mm through-hole	spare-I/O breakout, 46 signals + 4 GND	1	footprint only, not fitted by default
J8	USB-C receptacle, 16-pin (USB 2.0)	e.g. TYPE-C-31-M-12; CC1/CC2 → 5.1 kΩ Rd to GND	1	power + console on one cable
J9	DC barrel jack 5.5/2.1 mm	centre-positive, 5–12 V	1	CUI PJ-002AH or equivalent
J10	1×4 0.1" header	3V3, UART_TX(F), UART_RX(F), GND	1	direct console access if you do not want to fit the CH340C (0 Ω links R21–R24 select, §8)
SW1	SKQG tactile switch	reset → ext_rst_n (pin 134, drive to GND)	1	

Ref	Part number	Specification	Qty	Notes
SW2	SKQG tactile switch	PROGRAMN to GND (reconfigure)	1	
JP1	2-pin 0.1" jumper	disables Q1 (DTR auto-program)	1	
TP1-TP6	test points (1 mm pad or loop)	3V3, 2V5, 1V1, GND, DONE, INITN	6	bring-up / scope access

3.3 Passives and crystals

Ref	Value / type	Package	Qty	Purpose
R1	10 k Ω	0402	1	ext_rst_n pull-up (input has internal pull-up; the external resistor is for noise and SW1)
R2	4.7 k Ω	0402	1	PROGRAMN pull-up to VCCIO8 (TN-02038 Table 6.2)
R3	4.7 k Ω	0402	1	INITN pull-up to VCCIO8
R4	4.7 k Ω	0402	1	DONE pull-up to VCCIO8 (open-drain output)
R5	4.7 k Ω	0402	1	CFG1 = 1 strap to VCCIO8
R6, R7	1 k Ω	0402	2	CFG0 = 0 and CFG2 = 0 straps to GND (Master SPI = 010)
R8	1 k Ω	0402	1	MCLK/CCLK pull-up to VCCIO8 (hardware checklist)
R9-R12	100 Ω	0402	4	series resistors in the four Master-SPI nets (CCLK, DI, DO, CSSPIN) — damping + limits any contention with GPIOA[1]/[2]/[4]/[6] (§6.4)
R13-R15	10 k Ω	0402	3	config-flash (U2) /CS, /WP, /HOLD pull-ups to 3V3 — /CS must stay high when the configuration port floats
R16-R19	10 k Ω	0402	4	application-flash (U3) /CS, /WP, /HOLD pull-ups; motor_fault_n pull-up
R21-R24	0 Ω	0402	4	UART routing select: R21/R23 = CH340C, R22/R24 = J10 (populate one pair only)

Ref	Value / type	Package	Qty	Purpose
R25	4.7 kΩ	0402	1	Q1 gate resistor (DTR auto-program)
R26	10 kΩ	0402	1	Q1 gate pull-down (keeps PROGRAMN released when the jumper is open)
R27–R30	470 Ω	0402	4	user LED series resistors (D2–D5), 3 mA at 3.3 V – 2.0 Vf
R31–R34	1 kΩ	0402	4	indicator drives: R31 = Q4 base (INITN), R32 = Q2 base (DONE), R33 = D1 series, R34 = D6 series
R36, R37	5.1 kΩ	0402	2	USB-C CC1/CC2 Rd (required: tells the host to supply 5 V)
C1	100 nF	0402	1	ext_rst_n RC (with R1)
C2–C7	100 nF	0402	6	VCC (1.1 V) decoupling, one per VCC pin (pins 20, 29, 38, 66, 83, 130)
C8–C11	100 nF	0402	4	VCCAUX (2.5 V) decoupling, one per VCCAUX pin (17, 53, 96, 132)
C12–C20	100 nF	0402	9	VCCIO decoupling, one per VCCIO pin (9, 16, 36, 43, 70, 86, 100, 122, 137)
C21	4.7 μF	0603 X5R	1	1V1 bulk
C22	4.7 μF	0603 X5R	1	2V5 bulk
C23, C24	22 μF	0805 X5R	2	AP63203 input/output bulk
C25	4.7 μF	0603 X5R	1	3V3 bulk near banks 0/1/6/7
C26	100 nF	0402	1	Y1 oscillator supply decoupling
C27	100 nF	0402	1	CH340C supply decoupling
C28, C29	100 nF	0402	2	flash decoupling (one per device)
L1	4.7 μH / 2 A shielded	4×4 mm	1	AP63203 inductor (vendor reference-design value — re-check at layout)
L2	2.2 μH / 3 A shielded	4×4 mm	1	TPS62823 inductor (vendor reference-design value)

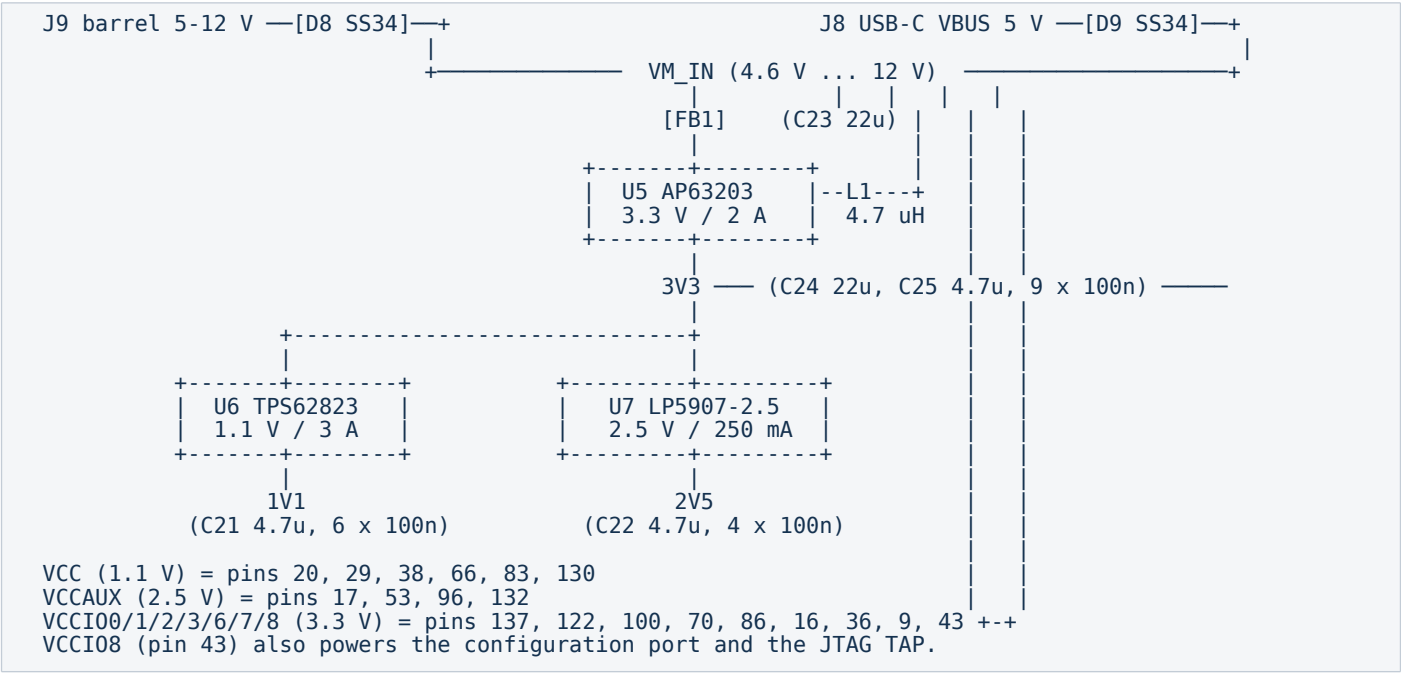
Ref	Value / type	Package	Qty	Purpose
FB1	ferrite bead 600 Ω @ 100 MHz	0603	1	VM_IN $\rightarrow$ analog side of the input, cuts motor/USB noise

Cost band (1-off, indicative): U1 USD 14-25 dominates; everything else together is roughly USD 6-10. A W25Q32JV + AP63203 + CH340C configuration lands near the bottom of that band.

Feed-through / mechanical (not on the PCB): 4 $\times$ M3 mounting holes on a 60 $\times$ 60 mm outline, optional 3.3 V FT232H or FT2232H adapter for JTAG, USB-C cable, 5-12 V supply.

4. Power architecture

4.1 Rail generation



4.2 Rails and current budget

Rail	Pins (count)	Nominal	Allowed range (datasheet)	Consumers	Static budget
1V1 (VCC)	20, 29, 38, 66, 83, 130 (6)	1.10 V	1.045-1.155 V (Table 3.2); abs max 1.32 V (Table 3.1)	FPGA core	77 mA typical (Table 3.8, T _J = 85 °C) + switching current; regulator rated 3 A
2V5 (VCCAUX)	17, 53, 96, 132 (4)	2.50 V	2.375-2.625 V; abs max 2.75 V	differential/referenced input buffers	16 mA typical (Table 3.8); regulator 250 mA
3V3 (VCCIO + I/O)	9, 16, 36, 43, 70, 86, 100, 122, 137 (9)	3.30 V	3.135-3.465 V for LVCMOS33 (Table 3.11); abs max 3.63 V	6 banks + 2 flashes + oscillator + LEDs + CH340C	6 $\times$ 0.5 mA typical (Table 3.8) + loads; regulator 2 A
VM_IN	J9 / J8	5-12 V	AP63203 input 3.8-32 V	U5, J2/J3 5 V pins, J6 motor passthrough	external supply must deliver $\geq$ 1 A

Timing margin is not the same thing as power margin. The static numbers above are from the datasheet; this repository has **no** dynamic power estimate for the SoC (the open toolchain has no power analyser, and `nextpnr` emits none). The regulators are therefore specified with a large margin, and "measure the real 1V1 current during bring-up" is an explicit bring-up step (§10) and an open item (§13.2 OQ-B1). Nothing on the board draws more than 3 A on any rail even in the worst case, and the 1V1 buck is current-limited, so an over-estimate cannot damage the device.

4.3 Power-up order, POR and the rules the board must not break

Rule	Requirement	How this board satisfies it
Recommended levels	VCC 1.045–1.155 V, VCCAUX 2.375–2.625 V, VCCIO 1.14–3.465 V	fixed-value regulators: 1.10 V / 2.50 V / 3.30 V
Ramp rate	all supplies 0.01–10 V/ms; VCCAUX $\leq$ 30 mV/ μ s	AP63203 $\approx$ 1 V/ms soft-start, TPS62823 $\approx$ 0.5 V/ms, LP5907 $\approx$ 0.1 V/ms (all inside the window)
POR release	only when VCC, VCCAUX and VCCIO8 are all above their trip points (0.90–1.00 V, 2.00–2.20 V, 0.95–1.06 V)	all three rails are generated and all three are monitored by the device itself; no board-side reset generator is needed
Master-SPI order (§3.5)	ramp VCCIO8 above the flash VIH before VCC or VCCAUX reach their POR trip, or hold PROGRAMN/INITN low until VCCIO8 is valid	1V1 and 2V5 are derived from 3V3, so 3V3 (= VCCIO8) is always valid first; the second mechanism (PROGRAMN held low) is available as a belt-and-braces option by fitting R2 to a supervisory reset instead of a plain pull-up
VCCIO8 ramp-down	the device has no ramp-down detection on VCCIO8; a brown-out that only drops VCCIO8 while VCC stays valid can leave the part in an undefined state	the 3V3 rail is the master: if the supply collapses, VCC collapses with it (all rails come from the same VM_IN), so the device sees a clean power cycle. Do not add a separate 3V3-only cutoff switch
Sequencing on power-down	no requirement	bulk capacitance (C21–C25) is sized so all three rails fall together, VCCIO8 last, because it has the largest load
JTAG VREF	the 1×10 Versa-style header has no VREF pin, the 2×5 header does	J1.1 and J2.1 both sit on 3V3; adapters that measure VREF will read 3.3 V, and adapters that level-shift from VREF will drive 3.3 V logic correctly

4.4 Decoupling

Rail	Bulk	Per-pin ceramic	Placement rule
1V1	C21 4.7 μ F	C2–C7 100 nF (6, one per VCC pin)	100 nF within 3 mm of each pin, via to the ground plane directly beside the pad
2V5	C22 4.7 μ F	C8–C11 100 nF (4, one per VCCAUX pin)	same
3V3	C24 22 μ F + C25 4.7 μ F	C12–C20 100 nF (9, one per VCCIO pin)	same; the 100 nF on VCCIO8 (pin 43) is the most important one on the board — it also decouples the configuration port
Input	C23 22 μ F	—	at the input diodes

Total ceramic count $\approx$ 40 pieces of 100 nF — no exotic parts, no bulk electrolytics.

5. Clocking

```
Y1 (25.000 MHz CMOS X0, 3.3 V) — CLK25 → U1 pin 133 clk_25m
| power: 3V3 + C26 100 nF
| output enable: tied to 3V3 (always on)
```

Item	Value	Source
Frequency	25.000 MHz	constraints/ecp5_144tqfp.lpf (clk_25m), docs/RESET_AND_CLOCK.md
Stability	±50 ppm is enough: the console runs 115200 8N1 ($\approx \pm 2\%$ tolerance) and the ISA/timer tests are cycle-based, not frequency-based	this design
PLL configuration	CLKI_DIV 2, CLKFB_DIV 3, CLKOP_DIV 16 → CLKOP = 37.5 MHz, VCO = 600 MHz (inside the 400-800 MHz range)	rtl/sv16_pll.sv, docs/ARCHITECTURE_DECISIONS.md ADR-024
Achieved timing	45.45 MHz Fmax at the 37.5 MHz constraint (build/pll/sv16_nextpnr.log), 44.70 MHz in the CLKSRC=osc 25 MHz build	docs/SYNTHESIS_AND_DEPLOYMENT.md
Second clock	none. No DDR, no SERDES (the ECP5 144-TQFP offering has 0 SerDes channels), no second oscillator input — the design has a single clock domain plus the hardened blocks	datasheet Table 1.1 / §4.3.2
JTAG clock	TCK ≤ 25 MHz (TAP_SCAN_CLOCK of TCK is (25.0e6, BOTH) in the BSDL model)	FPGA-MD-02097
Configuration clock	25 MHz pin 54 output in Master SPI; the device starts at ≈ 2.4 MHz and may step up per the bitstream's MCCLK_FREQ. A 293 KB compressed image therefore takes ≈ 1 s to load	TN-02039 §6.1, estimate

6. FPGA pin map and the board nets

board/TQFP144_PINOUT.md is the complete, generated 144-pin table (device pad, bank, rail, design signal, drive strength, pull mode and board destination). This section is the human-readable summary.

6.1 Signals used by the current design (52 of 98 I/O pins)

Function	Bank	Pins (TQFP144)	Drive / pull (from the LPF)	Board destination
25 MHz clock	0	133	4 mA	Y1.3
Reset button input	0	134	4 mA / pull-up	SW1 + R1 + C1
Console UART	3	73 (rx), 74 (tx)	4 mA	CH340C (U4) / J10
Application flash (bank-1 SPI)	1	110 sck, 111 cs_n, 112 mosi, 113 miso	8/8/8/4 mA	U3 (application flash)
SPI0 expansion	1	114 sck, 115 cs_n, 116 mosi, 117 miso	8/8/8/4 mA	J3 (EXP-A)
Status LEDs (mirror of GPIOA[3:0])	8	39, 40, 41, 44	8 mA	D2-D5 + R27-R30

Function	Bank	Pins (TQFP144)	Drive / pull (from the LPF)	Board destination
GPIOA[0:7]	8	45-52	8 mA	J5 EXP-C + config-flash nets for [1],[2],[4],[6] (§6.4)
GPIOA[8:15]	2	97, 98, 99, 104, 105, 106, 107, 108	8 mA	J3 EXP-A
GPIOB[0:7]	0	135, 136, 139, 140, 141, 142, 143, 128	8 mA	J4 EXP-B
GPIOB[8:11]	1	124, 125, 126, 127	8 mA	J4 EXP-B
GPIOB[12:15]	7	1, 2, 3, 4	8 mA	J4 EXP-B
PWM / motor	2	88 pwm (16 mA), 89 dir1, 102 dir2, 103 fault_n (pull-up)	see left	J6 motor header

LEDs are not independent software GPIO: `rtl/sv16_top.sv` does `assign led = ~gpio_a_out[3:0]`, so the LED pins always mirror GPIOA[3:0] and light when the bit is **0** (active low). GPIOA[3:0] are also on J5 pins 4/5/6/7 — the header copy and the LED copy are separate FPGA pins, so a lever on J5 does not have to disturb the LED indication, but both reflect the same register bit.

6.2 Spare I/O (46 pins)

Bank	Free pins	Where brought
1	118, 119, 120, 121	J7
2	90, 91, 92, 93, 94, 95	J7
3	67, 68, 69, 71, 72, 76, 77, 78, 79, 80, 81, 82, 84	J7
6	18, 19, 22, 23, 24, 25, 26, 27, 28, 30, 31, 33, 34, 35, 37	J7
7	5, 6, 7, 10, 11, 12, 13, 14	J7

Banks 6 and 7 are unused by the current bitstream, so adding a peripheral to them costs nothing in LUTs today. Two notes for whoever uses them: bank 3 pin 79 (PR35B, VREF1_3), bank 6 pin 19 (PL35B, VREF1_6) and bank 7 pin 11 (PL14C, VREF1_7) are VREF inputs, and bank 2 pin 88 is the 16 mA PWM output — do not stack heavy loads on it (it is already the motor PWM).

6.3 Active-low and direction conventions

Signal	Direction	Idle / reset state	Consequence
ext_rst_n	input	high (pull-up), press = low	SW1 resets the SoC; the startup FSM then runs the boot sequence
led[3:0]	output	LEDs off (GPIOA[3:0] = 1 after reset)	active low
flash_cs_n, spi0_cs_n	output	high = deselected	active low
motor_fault_n	input	pulled up = "no fault"	active low; the motor block at 0xF030 reports it (docs/PERIPHERALS.md)

Signal	Direction	Idle / reset state	Consequence
uart_rx	input	pulled up; must not be held low at power-up	the startup FSM forces the monitor after 4096 clocks of a held-low RX (RX_HOLD_CYCLES) — deliberate, and the reason the CH340C's TXD idles high
all GPIO	bidirectional	dir_reg = 0 on reset ⇒ inputs (high-Z)	this is what makes §6.4 safe in software

6.4 The one hard constraint on this board: the Master-SPI pads are also GPIOA pins

The design constrains `gpio_a[0:7]` to bank-8 pins 45–52. Four of those pads are the device's Master-SPI configuration port:

TQFP pin	Pad	sysCONFIG role	Design signal	Board net
46	PB11A	D1/MISO — input from the config flash	<code>gpio_a[1]</code>	J5.2 and U2.2 (flash DO) via R11 100 Ω
47	PB11B	D0/MOSI — output to the config flash	<code>gpio_a[2]</code>	J5.3 and U2.5 (flash DI) via R10 100 Ω
49	PB15A	HOLDN/DI/BUSY/CSSPIN — flash chip select	<code>gpio_a[4]</code>	J5.5 and U2.1 (/CS) via R12 100 Ω
51	PB15B	DOUT/CS0N — unused in a single-FPGA chain	<code>gpio_a[6]</code>	J5.7 only (no flash connection)
39, 40, 41, 44, 45, 48, 50, 52	PB6A, PB4A, PB6B, PB4B, PB9A, PB13A, PB13B, PB18A	parallel / slave-SPI pads, not claimed by Master SPI	<code>led[0:3]</code> , <code>gpio_a[0]</code> , <code>[3]</code> , <code>[5]</code> , <code>[7]</code>	headers / LEDs, free

Why this is nevertheless safe, and what must be true for it to stay safe:

1. During configuration the configuration logic drives pins 54, 47, 46 and 49 and the fabric is held in reset, so the user I/O on those pads cannot fight it.
2. After configuration, the pads are released to the fabric **because the design does not set the PERSISTENT attribute** on the spi port. This is what "recovering the configuration port pins for use as general purpose I/O" means in TN-02039 §4.6.10. Keeping it that way is deliberate.
3. The **board uses a separate flash for firmware** (U3 on bank-1 pins 110–113). The bitstream flash (U2) is only touched by the configuration logic and by JTAG, never by the soft-core.
4. The four shared nets carry 100 Ω series resistors (R9–R12). If firmware ever drives `gpio_a[1]`, `[2]`, `[4]` or `[6]` as an output while U2 is driving DO, the series resistor bounds the current instead of letting two drivers fight. Software must still treat those four bits as **inputs** (they are inputs after reset, and nothing in the ROM monitor or the bootloader ever changes that).
5. Consequence for a user: J5 pins 2, 3, 5 and 7 are not free general-purpose pins on this board — they are the configuration flash's DO, DI, /CS and the unused DOUT line. Use a spare pin from §6.2 if you need a clean signal, or apply variant V1 (§14) which remaps those four GPIO bits.

Everything else about the pin map is uncomplicated: no pin used by the design is shared with the JTAG TAP, and no pin used by the design is one of the dedicated sysCONFIG pins (PROGRAMN 57, INITN 55, DONE 56, CCLK 54, CFG_0 62, CFG_1 59, CFG_2 58, TCK 63, TMS 64, TDI 61, TDO 60).

7. Connection circuit diagram

Sheets 1–5 below are the connection diagram. Net labels are the LPF signal names, so any net can be traced back to a line in `constraints/ecp5_144tqfp.lpf` and to a row of `board/TQFP144_PINOUT.md`. Pin numbers for U1 are TQFP-144 physical pins; pin numbers for the flashes are SOIC-8 as defined by the W25Q datasheet; the CH340C is drawn by pin name (the vendor's SOP-16 numbering applies).

7.1 Sheet 1 — FPGA supply and ground

1V1 rail	2V5 rail	3V3 rail
+---+---+---+---+---+---+ 20 29 38 66 83 130 VCC VCC VCC VCC VCC VCC +---+---+---+---+---+---+ C2 C3 C4 C5 C6 C7 100n (all to GND) + C21 4u7	+---+---+---+---+---+---+ 17 53 96 132 AUX AUX AUX AUX +---+---+---+---+---+---+ C8 C9 C10 C11 100n (all to GND) + C22 4u7	+---+---+---+---+---+---+ 9 16 36 43 70 86 100 122 137 I07 I06 I06 I08 I03 I03 I02 I01 I00 +---+---+---+---+---+---+ C12 C13 C20 100n (all to GND) + C24 22u + C25 4u7
Ground: 8 (VSSI07) 15 (VSSI06) 21 (VSS) 32 (VSSI06) 42 (VSSI08) 65 (VSS) 75 (VSSI03) 85 (VSS) 87 (VSSI03) 101 (VSSI02) 123 (VSSI01) 129 (VSSI00) 131 (VSS) 138 (VSSI00) -> every pin to the GND plane with its own via. Pins 109 and 144 are NC.		

7.2 Sheet 2 — Configuration mode, JTAG, reset and status

```
U1.57 PROGRAMN --+--[R2 4.7k]-- 3V3
+-- SW2 --- GND (press = reconfigure)
+-- J1.4 (header PROGRAMn)
+-- Q1 drain ; Q1 source -- GND ; Q1 gate --[R25 4.7k]-- JP1 -- U4 DTR#
Q1 gate --[R26 10k]-- GND
(auto-reconfigure from the host, disable by removing JP1)

U1.55 INITN -----+--[R3 4.7k]-- 3V3
+-- J1.10 (header INITn)
+--[R31 1k]-- Q4 base Q4 = MMBT3906 (PNP, optional indicator)
Q4 emitter -- 3V3 ; Q4 collector -- D6 --[R34 1k]-- GND
(D6 red lights when INITN is low = SRAM clear / config error)

U1.56 DONE -----+--[R4 4.7k]-- 3V3
+-- J1.9 (header DONE)
+--[R32 10k]-- Q2 base Q2 = MMBT3904 (NPN)
Q2 emitter -- GND ; Q2 collector -- D1 --[R33 1k]-- 3V3
(D1 green lights when DONE is high = configured / in user mode)

Mode straps (Master SPI, CFG[2:0] = 010, latched on the rising edge of INITN):
U1.62 CFG_0 --[R6 1k]-- GND
U1.59 CFG_1 --[R5 4.7k]-- 3V3
U1.58 CFG_2 --[R7 1k]-- GND

JTAG TAP (dedicated pins, VCCI08 domain):
U1.63 TCK -- J1.8 -- J2.5 J1: 1=3V3(PWR) 2=TD0 3=TDI 4=PROGRAMn 5=NC
U1.64 TMS -- J1.6 -- J2.2 6=TMS 7=GND 8=TCK 9=DONE 10=INITn
U1.61 TDI -- J1.3 -- J2.3 J2: 1=3V3(VREF) 2=TMS 3=TDI 4=TD0 5=TCK
U1.60 TD0 -- J1.2 -- J2.4 6=GND 7..9=NC 10=GND

JTAG works with a blank or corrupt configuration flash -> this is the recovery path.
```

7.3 Sheet 3 — The two SPI flashes

```
U2 = W25Q64JVSSIQ – FPGA bitstream flash (Master SPI, driven by the configuration logic)
-----
U1.54 CCLK --[R9 100R]-----> U2.6 CLK ; U1.54 --[R8 1k]-- 3V3
U1.47 PB11B (MOSI pad) --[R10 100R]--> U2.5 DI ; also J5.3 as gpio_a[2] *
U1.46 PB11A (MISO pad) --[R11 100R]--> U2.2 DO ; also J5.2 as gpio_a[1] *
U1.49 CSSPIN pad --[R12 100R]--> U2.1 /CS ; also J5.5 as gpio_a[4] *
U2.1 --[R13 10k]-- 3V3 (deselect when idle)
U2.3 /WP --[R14 10k]-- 3V3
U2.7 /HOLD --[R15 10k]-- 3V3 (I02/I03 not used: no quad on TQFP144)
U2.8 VCC = 3V3 + C28 100n ; U2.4 GND

* the 100R series resistors bound any contention between the pads and the flash; firmware keeps
gpio_a[1]/[2]/[4]/[6] as inputs (see §6.4)

U3 = W25Q64JVSSIQ – SV-16 application flash (firmware slots, driven by the soft-core)
-----
U1.110 flash_sck -----> U3.6 CLK
U1.111 flash_cs_n -----> U3.1 /CS ; U3.1 --[R16 10k]-- 3V3
U1.112 flash_mosi -----> U3.5 DI
U1.113 flash_miso <----- U3.2 DO
U3.3 /WP --[R17 10k]-- 3V3
U3.7 /HOLD --[R18 10k]-- 3V3
U3.8 VCC = 3V3 + C29 100n ; U3.4 GND

Why two devices: the configuration port is not memory-mapped, so the soft-core cannot reach U2.
U3 is an ordinary SPI device on user I/O, which is what sv16_flash_ctrl (MMIO 0xF0A0) drives and what
the bootloader reads firmware images from. See §6.4 for the full argument.
```

7.4 Sheet 4 — Clock, reset and console

```
Y1 25.000 MHz X0 (3.3 V CMOS)          U4 CH340C (SOP-16, 3.3 V)
-----
Y1.3 OUT ----> U1.133 clk_25m          U4 VCC -- 3V3 + C27 100n ; U4 V3 -- 3V3 + 100n **
Y1.1 EN -- 3V3                          U4 GND -- GND
Y1.4 VCC -- 3V3 + C26 100n             U4 UD+ <--> J8 D+ (via USBLC6 D+ channel)
Y1.2 GND -- GND                        U4 UD- <--> J8 D- (via USBLC6 D- channel)
SW1 reset                             U4 TXD --[R21 0R]-- U1.73 uart_rx (idles high)
-----
U1.134 ext_rst_n --- SW1 -- GND         U4 RXD --[R23 0R]-- U1.74 uart_tx
      +--[R1 10k]-- 3V3                U4 DTR# -- JP1 -- Q1 gate (optional auto-reconfigure, §7.2)
      +-- C1 100n -- GND
U1.134 has PULLMODE=UP in the LPF; R1/C1 are for noise immunity and contact bounce,
not for correct reset behaviour.

J10 alternative console (populate R22/R24 instead of R21/R23)
-----
J10.1 3V3 | J10.2 UART_TX(F) --[R24 0R]-- U1.74 | J10.3 UART_RX(F) --[R22 0R]-- U1.73 | J10.4 GND
** 3.3 V-only board: a 5 V USB-UART adapter must not be used unless a level shifter is fitted
   (BOM "optional" row) or the adapter's I/O is 3.3 V tolerant.
```

7.5 Sheet 5 — LEDs, expansion and motor

```
Status LEDs (mirror of GPIOA[3:0], active low -- see §6.1)
-----
U1.39 led[0] --[R27 470R]-- D2 -- 3V3      (D2 on when GPIOA[0] = 0)
U1.40 led[1] --[R28 470R]-- D3 -- 3V3
U1.41 led[2] --[R29 470R]-- D4 -- 3V3
U1.44 led[3] --[R30 470R]-- D5 -- 3V3

Expansion headers (every signal is 3.3 V LVCMOS, 8 mA drive as constrained in the LPF)
-----
J5 EXP-C : 1 A0  2 A1* 3 A2* 4 A3  5 A4* 6 A5  7 A6* 8 A7  9 GND 10 3V3
J3 EXP-A : 1 3V3 2 GND 3 A8  4 A9  5 A10 6 A11 7 A12 8 A13 9 A14 10 A15
          11 spi0_cs_n 12 spi0_sck 13 spi0_mosi 14 spi0_miso
          15 GND 16 GND 17 5V 18 5V 19 GND 20 GND
J4 EXP-B : 1 3V3 2 GND 3 B0  4 B1  5 B2  6 B3  7 B4  8 B5  9 B6 10 B7
          11 B8 12 B9 13 B10 14 B11 15 B12 16 B13 17 B14 18 B15 19 GND 20 5V
J7 spare : 2x25, 46 spare I/O (bank 1 : 118-121, bank 2 : 90-95, bank 3 : 67,68,69,71,72,
          76-82,84, bank 6 : 18,19,22-28,30,31,33-35,37, bank 7 : 5-7,10-14) + 4 GND.
          Footprint only, not fitted.
* A1/A2/A4/A6 are the shared configuration-flash nets of §6.4; A3/A5/A7 are free.
The 5 V pins are USB VBUS only (the 3V3 rail comes from VM_IN, §4.1).

Motor connector (keyed 1x6)
-----
J6.1 VM <-- VM_IN through the keyed header (fused on the driver board, not on this board)
J6.2 GND -- GND
J6.3 PWM <----- U1.88 pwm_out (16 mA drive, PWM block at MMIO 0xF030)
J6.4 DIR1 <----- U1.89 motor_dir1 (mirror of GPIOA[4])
J6.5 DIR2 <----- U1.102 motor_dir2 (mirror of GPIOA[5])
J6.6 FAULTn <-----> U1.103 motor_fault_n (input, PULLMODE=UP + R19 10k pull-up to 3V3)
```

8. Connector pinouts and how to use them

Connector	Type	Pinout	Use
J1	1×10, 2.54 mm	1 3V3 · 2 TDO · 3 TDI · 4 PROGRAMn · 5 NC · 6 TMS · 7 GND · 8 TCK · 9 DONE · 10 INITn	Versa-style programming header, works with openFPGALoader and any FT232H/FT2232H adapter
J2	2×5, 1.27 mm keyed	1 3V3 (VREF) · 2 TMS · 3 TDI · 4 TDO · 5 TCK · 6 GND · 7-9 NC · 10 GND	Digilent/HS2-style probes; wired in parallel with J1 — use one at a time
J3	2×10, 2.54 mm	see §7.5	SPI0 + GPIOA[15:8]; SPI0 is the SoC's second SPI (0xF050), ideal for an SD card or radio module
J4	2×10, 2.54 mm	see §7.5	all 16 GPIOB pins (0xF070)

Connector	Type	Pinout	Use
J5	1×10, 2.54 mm	see §7.5	GPIOA[7:0]; pins 2/3/5/7 share the configuration flash (§6.4)
J6	1×6 keyed	see §7.5	motor driver interface
J7	2×25, 2.54 mm	46 spare I/O + 4 GND	expansion only; not fitted
J8	USB-C receptacle	VBUS, D+, D−, GND, CC1/CC2 = 5.1 kΩ Rd	power + console on one cable
J9	5.5/2.1 mm barrel	centre positive, 5–12 V	bench supply, powers the motor connector too
J10	1×4, 2.54 mm	3V3 · UART_TX(F) · UART_RX(F) · GND	console without fitting the CH340C (fit R22/R24, remove R21/R23)

Wiring notes that prevent the two most common support cases:

- **A 5 V USB-UART adapter on J10 will damage the FPGA pins** (they are LVCMOS33 with no 5 V tolerance documented for this configuration). Use 3.3 V levels, or fit the optional level shifter.
- **Two JTAG adapters on J1 and J2 at the same time** will fight. Populate one connector if you are unsure; they are on the same four nets.
- The **console is the only update path in the field**: bitstreams go through JTAG (or through U2), while firmware images go through the UART into U3. That asymmetry is inherent to soft-core + configuration flash and is worth putting on the silkscreen.

9. What lives in which memory, and how each part gets programmed

Storage	Device	Content	Written by	Survives power cycle
Configuration SRAM (inside U1)	—	the FPGA bitstream	configuration logic from U2, or JTAG directly	no
U2 bitstream flash	W25Q64JV	<code>build/pll/sv16_top.bit</code> (293 KB) at offset 0	<code>openFPGALoader -f</code> through the FPGA's MSPI port, or a bench SPI programmer (SOIC-8 clip)	yes
U3 application flash	W25Q64JV (W25Q40-class is enough, see below)	firmware images: slot A at 0x000000, slot B at 0x008000, 32 KB each, 32-byte header, slot record at bytes 0x18/0x19	the SV-16 ROM monitor over the UART (make <code>upload</code> , make <code>upload-slot</code>), or an external SPI programmer	yes
Boot ROM	hardened in the FPGA (initialised BRAM, .ORG 0xE000)	the ROM monitor (790 lines of SV-16 assembly)	bitstream build (make <code>rom</code>)	n/a (part of U2)
SRAM	FPGA block RAM	32 KB at 0x0000-0x3FFF — applications at run time, loader while booting	boot loader / firmware	no

`docs/BOOT_AND_PROGRAMMING.md` documents the image format, the A/B slot policy (PENDING → TRIED → GOOD or BAD), and the monitor's command set; this board only supplies the two physical flash devices it assumes. The typical application flash in that document is a **512 KB W25Q40-class part**, so U3 can be downsized to

W25Q40CLSNIQ (or W25Q16JV) without changing a line of firmware — the design only needs the standard SPI-NOR command set (0x03/0x0B read, 0x02/0x06/0x05/0x20/0xD8 program/erase/status) that the whole W25Q family shares. U2 must be at least as large as the bitstream: 1 MB is comfortable, 8 MB is what the BOM lists because the part is the same and cheaper in volume.

9.1 Programming flows on this board

```
# 1. Volatile configuration (fastest loop while developing the FPGA side)
make bitstream          # -> build/sv16_top.bit (J1/J2 + any FT232H/FT2232H adapter)
make prog CABLE=ft232 CABLE=... # openFPGALoader --fpga-part LFE5U-12F-6TG144C build/sv16_top.bit

# 2. Persistent bitstream (survives power cycles, loads from U2 in ~1 s at the default MCLK)
openFPGALoader -c ft232 -f build/sv16_top.bit # writes U2 through the FPGA's MSPI port

# 3. Firmware: single-image upload through the monitor (no rollback)
make upload PORT=/dev/ttyUSB0 IMG=build/fw/app_img.hex

# 4. Firmware: A/B field update (recommended -- a bad image cannot brick the board)
make slot-image SLOT=1 && make upload-slot SLOT=1 PORT=/dev/ttyUSB0
make mon-boot # boot the PENDING slot; loader marks it TRIED
make commit # once the new image behaves: record GOOD, retire the other slot
```

The JTAG headers stay useful even with a blank U2: openFPGALoader --detect should identify the device (the ECP5 12F IDCODE is device specific — record what your part reports; the 25F model in the BSDL FPGA-MD-02097 reads 0x41111043), then flow 1 or 2 recovers the board.

10. Bring-up and acceptance test plan

#	Step	Pass criterion	Tool
1	Resistance check before power	> 100 Ω from each rail (TP1/TP2/TP3) to GND; no short between rails	DMM
2	Power up on the barrel jack only	TP1 = 3.135–3.465 V, TP2 = 2.375–2.625 V, TP3 = 1.045–1.155 V	DMM
3	Ramp order (scope, two channels)	3V3 reaches ≈ 0.9 V before 1V1/2V5 start; no rail slower than 0.01 V/ms; VCCAUX slew ≤ 30 mV/ μ s	scope, 100 ms/div
4	JTAG detection	openFPGALoader --detect returns the ECP5 IDCODE, no cable/chain errors	J1/J2 + adapter
5	Volatile configuration	DONE (TP5) goes high, D1 lights, INITN stays high	make prog
6	Console	115200 8N1 shows the monitor banner; ? lists commands	make mon-term PORT=/dev/ttyUSB0
7	Reset and reset cause	SW1 restarts the monitor; SYS_RSTCAUSE (0xF003) reports the expected source	monitor
8	LEDs	an application that toggles GPIOA[3:0] blinks D2–D5 in the same pattern (active low)	any firmware/apps/ build
9	Application flash	monitor reads the JEDEC ID of U3; make upload writes and V verifies	monitor

#	Step	Pass criterion	Tool
10	A/B update	make upload-slot SLOT=1 + make mon-boot + make commit, then power-cycle and confirm the new image runs from slot B	host scripts
11	Rollback	upload a deliberately broken image to the free slot, boot it, reset — the loader must fall back to the other slot and set BOOT_ERR[7]	host scripts
12	Motor	scope on J6.3 shows the PWM waveform with the duty written to 0xF030; J6.4/J6.5 follow GPIOA[4]/[5]; pulling J6.6 low sets the fault bit	scope + firmware/apps/motor_test.s
13	Current measurement	record 1V1/2V5/3V3 current at 37.5 MHz with the monitor idle and with an application running (this closes OQ-B1)	bench supply
14	Long-run soak	24 h with the monitor running and the motor app cycling: no watchdog resets beyond those expected, no configuration re-entry	host log

11. Layout and EMC guidance

Topic	Guidance
Stack-up	4 layers, 1.6 mm: signal / solid GND / solid PWR (3V3, 1V1, 2V5 pours) / signal. A 2-layer board is possible but the 0.5 mm-pitch TQFP fan-out and the three rails make it unpleasant; not recommended
Fan-out	0.25 mm traces, 0.2 mm vias; TQFP-144 escape on all four sides. Ground pins 8/15/21/32/42/65/75/85/87/101/123/129/131/138 each get their own via to the GND plane
Decoupling	the 100 nF of C2–C20 within 3 mm of its pin, on the same side as the FPGA, with the via first at the pad; bulk (C21–C25) near the regulator outputs
Regulators	keep the two switchers' inductors and the input loop tight; keep the 1V1 inductor away from the configuration flash and the 25 MHz oscillator; put U5/U6/U7 on the opposite edge of the board from J6
Motor return	J6 GND must have its own path to the input bulk capacitor; do not route motor current through the FPGA ground area. FB1 and the 3V3 bulk capacitor are the barrier
Oscillator	Y1 within 10 mm of pin 133, no via in the trace, ground pour under the part, keep the switching nodes > 5 mm away
Config flash	U2 within 15 mm of pins 46/47/49/54; the 100 Ω series resistors at the FPGA end (they are there to bound contention, §6.4)
USB	D+/D– as a 90 Ω differential pair, ESD device at the connector, CC1/CC2 5.1 k Ω within 10 mm of J8

Topic	Guidance
JTAG	short TCK stub, no stubs beyond the two headers (TCK is the fastest signal at ≤ 25 MHz but the slowest is the cable); keep the TDO trace ≤ 50 mm
Test points	TP1-TP6 on the top side, labelled with the net names used in this document
Thermal	TQFP-144 has no exposed pad; use a copper pour on all four sides under the part, thermal vias under the body. Expected dissipation is well under 1 W (77 mA $\times$ 1.1 V core static plus switching), so no heatsink and no airflow requirement
Options/silkscreen	mark DNP parts (D6, J7, level shifter) and J5 pins 2/3/5/7 ("CFG FLASH — inputs only")

12. What this board turns the design into

MCU property	Where it comes from
Persistent program storage	U3 + sv16_flash_ctrl (MMIO 0xF0A0) + the bootloader reading A/B slots
Field reprogramming that cannot brick the board	A/B slots with PENDING/TRIED/GOOD records (ADR-019) + the monitor's upload/verify + JTAG as the last resort
Console / host interface	CH340C + UART0 at 115200 8N1, the same protocol scripts/sv16_mon.py speaks
Deterministic start-up	POR $\rightarrow$ configuration $\rightarrow$ startup FSM $\rightarrow$ 32768-cycle boot delay $\rightarrow$ boot attempt $\rightarrow$ monitor fallback (holding the RX line low for 4096 cycles forces the monitor, which is the escape hatch after a bad image)
Reset control with cause reporting	SW1 on ext_rst_n + SYS_RSTCAUSE at 0xF003
Autonomous safety	watchdog at 0xF080 (docs/PERIPHERALS.md)
Digital I/O with a real expansion story	46 spare I/O on J7, 32 GPIO on J3/J4/J5 (2 ports $\times$ 16 bits at 0xF010/0xF070), SPI0 at 0xF050
Motor/actuator control	PWM at 0xF030 (DD-16 mA output on pin 88) with direction pins and a fault input
Debug	ROM monitor over UART today; the roadmap item S3 (UART GDB stub $\rightarrow$ JTAGG bridge, REPORT.md §8.2) uses this same JTAG header

13. Verification status and open items

13.1 Verified (with the artefact that proves it)

Item	Evidence
Pin numbers, pads, banks of all 52 constrained signals	board/TQFP144_PINOUT.md, generated from the prjtrellis device database for LFE5U-12F and the Lattice BSDL PIN_MAP; scripts/sv16_board_pins.py --check fails on drift
Device pin budget (98 I/O, 11 dedicated, 6 VCC, 4 VCCAUX, 9 VCCIO, 14 GND, 2 NC = 144)	board/ecp5_tqfp144.json summary, validated by the script against both sources
Rail limits, ramp rates, POR trips, LVCMOS33 range	FPGA-DS-02012-3.4 Tables 3.1/3.2/3.3/3.4/3.5/3.11 (the PDF is in this repository)
Static supply currents	FPGA-DS-02012-3.4 Table 3.8

Item	Evidence
Configuration mode and pull-up values	TN1260 / TN-02039 (CFG[2:0] = 010) + TN-02038-2.0 Table 6.2 (4.7 kΩ PROGRAMN/INITN, 1 kΩ MCLK, 4.7–10 kΩ CSSPIN)
JTAG header pinout and TAP	ECP5 Versa EB98/EB103 Appendix A; BSDL TAP_SCAN_* attributes and BOUNDARY_LENGTH 409
Fit of the design in the device	build/pll/sv16_nextpnr.log: 9,191 LUT4, 4,772 FF, 18 DP16KD, 1 EHXPPLL, 45.45 MHz

13.2 Open items (must be closed before or during layout)

ID	Item	Why it is open	Closing action
OQ-B1	Dynamic current on 1V1/2V5/3V3	the open-source flow has no power analyser; only datasheet static numbers exist	measure on the first board (\$10 step 13); the regulators are oversized on purpose
OQ-B2	Regulator passives (L1 4.7 μH, L2 2.2 μH, capacitor values, feedback dividers)	taken from the vendors' typical application circuits, not re-derived here	run the vendors' design tools / reference schematics at layout time
OQ-B3	Configuration time from U2 at the default MCLK	TN-02039 gives the mechanism but the bitstream's MCCLK_FREQ setting is not inspected by the open flow	measure INITN→DONE with a scope on the first board
OQ-B4	CH340C 3.3 V connection details (V3 pin, decoupling, DTR polarity)	pin numbers and the 3.3 V configuration must come from the WCH datasheet, which is not in this repository	download and check the WCH datasheet before schematic capture
OQ-B5	USB-C sink compliance (Rd values, no-PD behaviour)	standard practice, no USB-IF testing	bench-test with several hosts/cables
OQ-B6	J2 (2×5 0.05") pin map versus the actual probe in use	conventions differ slightly between Digilent/FT2232H clones	check against the chosen adapter's cable
OQ-B7	EMC / ESD pre-compliance (motor transients, USB)	no test house, no hardware	add the layout measures of §11 and re-assess when hardware exists
OQ-B8	Mechanical outline, mounting, connector placement, enclosure	not fixed yet	decide with the mechanical drawing at PCB start
OQ-B9	J5 pin sharing with the configuration flash (§6.4)	acceptable as documented, but it is a usability wart	either keep the firmware rule or apply variant V1 (§14)

14. Variants and the next revision

Variant	Change	Benefit	Cost
V1 — decouple J5 from the config flash	move gpio_a[1], [2], [4], [6] in constraints/ecp5_144tqfp.1p from bank 8 (pins 46/47/49/51) to four of the free bank-2 pins (90–95), then regenerate board/TQFP144_PINOUT.md	J5 becomes 10 fully usable GPIO; the config flash no longer needs series resistors or a firmware rule; closes OQ-B9	one LPF edit, a bitstream rebuild, doc regeneration. Constraint changes touch the pin map docs and §6.4 only

Variant	Change	Benefit	Cost
V2 — single-cable USB	replace the CH340C with an FT232H (channel A = JTAG, channel B = UART)	one cable does configuration and console; no JTAG adapter needed	+USD 4–6, QFN-56 handling, two interfaces to document
V3 — on-board motor driver	populate U9 = DRV8871/TB6612 on the J6 nets, with current sense into a spare ADC-less comparator pin	turn-key actuator board	needs a driver design + thermal review
V4 — mixed I/O rails	split one bank to 1.8 V (e.g. for an SD card in bank 3)	expands the peripheral set	LPF IOSTANDARD change, an extra regulator, and the level-shift question returns
V5 — USB-C PD / PoE input	PD sink controller requesting 9–12 V, or 802.3af front end	no barrel jack	BOM + certification work

The order I would do them: **V1 first** (it is cheap and removes the only awkward constraint in this document), then V2 when the JTAG cable becomes the bottleneck, then V3 when a real motor is on the bench.

15. Appendix A — electrical quick reference (LFE5U-12F, from FPGA-DS-02012-3.4)

Parameter	Value	Table
VCC recommended / absolute max	1.045–1.155 V / –0.5...1.32 V	3.1, 3.2
VCCAUX recommended / absolute max	2.375–2.625 V / –0.5...2.75 V	3.1, 3.2
VCCIO recommended / absolute max	1.14–3.465 V / –0.5...3.63 V	3.1, 3.2
LVC MOS33 VCCIO window	3.135–3.465 V	3.11
Supply ramp	0.01–10 V/ms, monotonic; VCCAUX ≤ 30 mV/μs	3.3
POR trip (VCC / VCCAUX / VCCIO8)	0.90–1.00 V / 2.00–2.20 V / 0.95–1.06 V; no VCCIO8 ramp-down detection	3.4
POR release condition	VCC and VCCAUX and VCCIO8 above trip	3.5
Static current, LFE5U-12F (T _J 85 °C, outputs tri-stated)	ICC 77 mA · ICCAUX 16 mA · ICCIO 0.5 mA/bank	3.8
Master SPI timing	fMCLK ±20 %, duty 40–60 %; PROGRAMN low ≥ 110 ns (≤ 50 ns is rejected); tINITL ≤ 55 μs; tVMC ≤ 5 μs; tCZ ≤ 300 ns	p85–87
JTAG TCK	≤ 25 MHz (BSDL TAP_SCAN_CLOCK)	FPGA-MD-02097
Junction temperature	0–85 °C (commercial C grade)	3.2

16. Appendix B — files this document depends on

File	Role
board/ecp5_tqfp144.json	device pin database (built from the prjtrellis database + Lattice BSDL)
board/TQFP144_PINOUT.md	generated 144-pin net table — the netlist appendix of this document

File	Role
scripts/sv16_board_pins.py	regenerates the table, checks it for drift, prints the bank summary
constraints/ecp5_144tqfp.lpf	the single source of truth for signal → pin (and for drive strength / pull mode)
docs/BOARD_AND_BRINGUP.md	bench notes, harness wiring and bring-up history for the simulation and previous hardware effort — read alongside §10
docs/SYNTHESIS_AND_DEPLOYMENT.md	build, timing, utilisation and existing pin-map narrative
docs/BOOT_AND_PROGRAMMING.md	image format, slot policy, monitor protocol, host tools
docs/MEMORY_MAP.md, docs/PERIPHERALS.md	MMIO map and register reference for every signal that reaches a connector
docs/MCU_READINESS.md	what is still missing for a production-grade MCU (the board does not change that list)
REPORT.md	status of the whole project, including the software-only roadmap §8.2
ECP5 and ECP5-5G.pdf	Lattice FPGA-DS-02012-3.4 — every rail, timing and pin-count number above

Revision 1.0 — first issue, written against `constraints/ecp5_144tqfp.lpf` and the bitstream in `build/pll/sv16_top.bit`. No board exists yet; this document is the specification to build one.

LFE5U-12F-6TG144C — complete TQFP-144 pin / net table

Generated by scripts/sv16_board_pins.py — do not edit by hand. Device data: board/ecp5_tqfp144.json; design signals: constraints/ecp5_144tqfp.lpf; board destinations: BOARD.md §5–§7. Part: **LFE5U-12F-6TG144C** (TQFP144 (20 x 20 mm, 0.5 mm pitch)).

1. I/O bank summary

Bank	VCCIO rail	I/O pins	Used by this design	Free
0	3V3 (pin 137)	10	10	0
1	3V3 (pin 122)	16	12	4
2	3V3 (pin 100)	18	12	6
3	3V3 (pin 70, 86)	15	2	13
6	3V3 (pin 16, 36)	15	0	15
7	3V3 (pin 9)	12	4	8
8	3V3 (pin 43)	12	12	0

Total: 98 user I/O, 52 used by the design, 46 free.

2. Configuration-pad interactions (CFG[2:0] = 010, Master SPI)

Pin	Pad	sysCONFIG role	LPF signal	Consequence on the board
39	PB6A	D5/IO5	led[0]	parallel/slave-mode pad only, free in this configuration
40	PB4A	D7/IO7	led[1]	parallel/slave-mode pad only, free in this configuration
41	PB6B	D4/IO4	led[2]	parallel/slave-mode pad only, free in this configuration
44	PB4B	D6/IO6	led[3]	parallel/slave-mode pad only, free in this configuration
45	PB9A	D3/IO3	gpio_a[0]	parallel/slave-mode pad only, free in this configuration
46	PB11A	D1/MISO (Master SPI pad)	gpio_a[1]	shares the net with the config flash — see BOARD.md §6.4
47	PB11B	D0/MOSI (Master SPI pad)	gpio_a[2]	shares the net with the config flash — see BOARD.md §6.4
48	PB13A	CSN/SN	gpio_a[3]	parallel/slave-mode pad only, free in this configuration
49	PB15A	HOLDN/DI/BUSY/CSSPIN (Master SPI pad)	gpio_a[4]	shares the net with the config flash — see BOARD.md §6.4

Pin	Pad	sysCONFIG role	LPF signal	Consequence on the board
50	PB13B	CS1N	gpio_a[5]	parallel/slave-mode pad only, free in this configuration
51	PB15B	DOUT/CSON (Master SPI pad)	gpio_a[6]	shares the net with the config flash — see BOARD.md §6.4
52	PB18A	WRITEN	gpio_a[7]	parallel/slave-mode pad only, free in this configuration

Pins sharing a net with the configuration flash: 46, 47, 49, 51.

3. Complete pin table (1-144)

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
1	PL5C	io	bank 7 (3V3)	gpio_b[12]	8 mA	J3.15 EXP-B header
2	PL5D	io	bank 7 (3V3)	gpio_b[13]	8 mA	J3.16 EXP-B header
3	PL5B	io	bank 7 (3V3)	gpio_b[14]	8 mA	J3.17 EXP-B header
4	PL8A	io	bank 7 (3V3)	gpio_b[15]	8 mA	J3.18 EXP-B header
5	PL8C	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
6	PL8B	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
7	PL8D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
8	VSSIO7	gnd	GND plane	—	—	GND plane
9	VCCIO7	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
10	PL14A	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
11	PL14C	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
12	PL14B	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
13	PL14D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
14	PL23D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
15	VSSIO6	gnd	GND plane	—	—	GND plane
16	VCCIO6	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
17	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
18	PL32B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
19	PL35B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
20	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
21	VSS	gnd	GND plane	—	—	GND plane
22	PL35C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
23	PL35D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
24	PL38A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
25	PL38B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
26	PL41C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
27	PL41D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
28	PL44A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
29	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
30	PL44B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
31	PL44C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
32	VSSIO6	gnd	GND plane	—	—	GND plane
33	PL44D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
34	PL47A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
35	PL47B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
36	VCCIO6	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
37	PL47C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
38	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
39	PB6A	io	bank 8 (3V3)	led[0]	8 mA	D2 LED (active low) + R31 470R
40	PB4A	io	bank 8 (3V3)	led[1]	8 mA	D3 LED (active low) + R32 470R
41	PB6B	io	bank 8 (3V3)	led[2]	8 mA	D4 LED (active low) + R33 470R
42	VSSIO8	gnd	GND plane	—	—	GND plane
43	VCCIO8	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
44	PB4B	io	bank 8 (3V3)	led[3]	8 mA	D5 LED (active low) + R34 470R
45	PB9A	io	bank 8 (3V3)	gpio_a[0]	8 mA	J4.1 EXP-C header
46	PB11A	io	bank 8 (3V3)	gpio_a[1]	8 mA	J4.2 EXP-C header
47	PB11B	io	bank 8 (3V3)	gpio_a[2]	8 mA	J4.3 EXP-C header
48	PB13A	io	bank 8 (3V3)	gpio_a[3]	8 mA	J4.4 EXP-C header
49	PB15A	io	bank 8 (3V3)	gpio_a[4]	8 mA	J4.5 EXP-C header
50	PB13B	io	bank 8 (3V3)	gpio_a[5]	8 mA	J4.6 EXP-C header
51	PB15B	io	bank 8 (3V3)	gpio_a[6]	8 mA	J4.7 EXP-C header
52	PB18A	io	bank 8 (3V3)	gpio_a[7]	8 mA	J4.8 EXP-C header
53	VCCAUX	vcclaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
54	CCLK	config	config	—	—	U5.6 config flash CLK (R7 100R series, R8 1k pull-up to VCCIO8)
55	INITN	config	config	—	—	J1.10 + R9 4.7k pull-up to VCCIO8 + D6 red LED via R15 1k
56	DONE	config	config	—	—	J1.9 + R10 4.7k pull-up to VCCIO8 + D1 green LED via R32 1k
57	PROGRAMN	config	config	—	—	J1.4 + R11 4.7k pull-up to VCCIO8 + SW2 + Q1 (DTR auto-program, JP1 to disable)
58	CFG_2	config	config	—	—	R14 1k to GND (CFGMDN2 = 0)
59	CFG_1	config	config	—	—	R13 4.7k to VCCIO8 (CFGMDN1 = 1)
60	TDO	jtag	JTAG	—	—	J1.2 (TDO)
61	TDI	jtag	JTAG	—	—	J1.3 (TDI)
62	CFG_0	config	config	—	—	R12 1k to GND (CFGMDN0 = 0)
63	TCK	jtag	JTAG	—	—	J1.8 (TCK)
64	TMS	jtag	JTAG	—	—	J1.6 (TMS)
65	VSS	gnd	GND plane	—	—	GND plane
66	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
67	PR47B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
68	PR47D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
69	PR47C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
70	VCCIO3	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
71	PR47A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
72	PR44C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
73	PR41D	io	bank 3 (3V3)	uart_rx	4 mA / UP	U1 TXD via R21 0R / J5.3 via R22 0R
74	PR41C	io	bank 3 (3V3)	uart_tx	4 mA	U1 RXD via R23 0R / J5.2 via R24 0R
75	VSSIO3	gnd	GND plane	—	—	GND plane
76	PR41B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
77	PR41A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
78	PR35D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
79	PR35B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
80	PR35C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
81	PR35A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
82	PR32D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
83	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
84	PR32C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
85	VSS	gnd	GND plane	—	—	GND plane
86	VCCIO3	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
87	VSSIO3	gnd	GND plane	—	—	GND plane
88	PR23D	io	bank 2 (3V3)	pwm_out	16 mA	J6.3 motor connector
89	PR23B	io	bank 2 (3V3)	motor_dir1	8 mA	J6.4 motor connector
90	PR23C	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
91	PR23A	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
92	PR20D	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
93	PR20B	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
94	PR20C	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
95	PR20A	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
96	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
97	PR14B	io	bank 2 (3V3)	gpio_a[8]	8 mA	J2.3 EXP-A header
98	PR14C	io	bank 2 (3V3)	gpio_a[9]	8 mA	J2.4 EXP-A header
99	PR14A	io	bank 2 (3V3)	gpio_a[10]	8 mA	J2.5 EXP-A header
100	VCCIO2	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
101	VSSIO2	gnd	GND plane	—	—	GND plane
102	PR8D	io	bank 2 (3V3)	motor_dir2	8 mA	J6.5 motor connector
103	PR8B	io	bank 2 (3V3)	motor_fault_n	4 mA / UP	J6.6 motor connector
104	PR8C	io	bank 2 (3V3)	gpio_a[11]	8 mA	J2.6 EXP-A header
105	PR8A	io	bank 2 (3V3)	gpio_a[12]	8 mA	J2.7 EXP-A header
106	PR2D	io	bank 2 (3V3)	gpio_a[13]	8 mA	J2.8 EXP-A header
107	PR2C	io	bank 2 (3V3)	gpio_a[14]	8 mA	J2.9 EXP-A header
108	PR2B	io	bank 2 (3V3)	gpio_a[15]	8 mA	J2.10 EXP-A header
109	NC	nc	do not connect	—	—	do not connect
110	PT67A	io	bank 1 (3V3)	flash_sck	8 mA	U6.6 application flash SCK
111	PT62B	io	bank 1 (3V3)	flash_cs_n	8 mA	U6.1 application flash /CS

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
112	PT62A	io	bank 1 (3V3)	flash_mosi	8 mA	U6.5 application flash DI
113	PT60B	io	bank 1 (3V3)	flash_miso	4 mA / UP	U6.2 application flash DO
114	PT60A	io	bank 1 (3V3)	spi0_sck	8 mA	J2.12 SPI0 expansion
115	PT58B	io	bank 1 (3V3)	spi0_cs_n	8 mA	J2.11 SPI0 expansion
116	PT58A	io	bank 1 (3V3)	spi0_mosi	8 mA	J2.13 SPI0 expansion
117	PT56B	io	bank 1 (3V3)	spi0_miso	4 mA / UP	J2.14 SPI0 expansion
118	PT56A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
119	PT53B	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
120	PT53A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
121	PT51A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
122	VCCIO1	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
123	VSSIO1	gnd	GND plane	—	—	GND plane
124	PT42B	io	bank 1 (3V3)	gpio_b[8]	8 mA	J3.11 EXP-B header
125	PT42A	io	bank 1 (3V3)	gpio_b[9]	8 mA	J3.12 EXP-B header
126	PT40B	io	bank 1 (3V3)	gpio_b[10]	8 mA	J3.13 EXP-B header
127	PT40A	io	bank 1 (3V3)	gpio_b[11]	8 mA	J3.14 EXP-B header
128	PT29A	io	bank 0 (3V3)	gpio_b[7]	8 mA	J3.10 EXP-B header
129	VSSIO0	gnd	GND plane	—	—	GND plane
130	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
131	VSS	gnd	GND plane	—	—	GND plane
132	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
133	PT15B	io	bank 0 (3V3)	clk_25m	4 mA	Y1.3 25 MHz XO output

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
134	PT15A	io	bank 0 (3V3)	ext_rst_n	4 mA / UP	SW1 reset button (to GND) + R1 10k pull-up + C1 100n
135	PT13B	io	bank 0 (3V3)	gpio_b[0]	8 mA	J3.3 EXP-B header
136	PT13A	io	bank 0 (3V3)	gpio_b[1]	8 mA	J3.4 EXP-B header
137	VCCIO0	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
138	VSSIO0	gnd	GND plane	—	—	GND plane
139	PT9B	io	bank 0 (3V3)	gpio_b[2]	8 mA	J3.5 EXP-B header
140	PT9A	io	bank 0 (3V3)	gpio_b[3]	8 mA	J3.6 EXP-B header
141	PT6B	io	bank 0 (3V3)	gpio_b[4]	8 mA	J3.7 EXP-B header
142	PT4B	io	bank 0 (3V3)	gpio_b[5]	8 mA	J3.8 EXP-B header
143	PT6A	io	bank 0 (3V3)	gpio_b[6]	8 mA	J3.9 EXP-B header
144	NC	nc	do not connect	—	—	do not connect

Pad names and the dedicated sysCONFIG/TAP pins are from Lattice BSDL FPGA-MD-02097 (LFE5U_25F_XXTG144); the LFE5U-12F/25F/45F 144-TQFP pinout is identical (FPGA-DS-02012-3.4 Table 1.1 lists 98 user I/O on 144 TQFP for each density). The 98 user-I/O pin numbers and their banks are taken from the prjtrellis database for LFE5U-12F and agree with the BSDL.